Paper and journal	Study Objective	Method Followed	Data Inputs Used	Modeling / Analysis Type	Key Outputs / Findings	Notes
Springer – *Environmental Science and Technology Library* Gibson J.M., Brammer A.S., Davidson C.A., Folley T., Launay F.J.P., Thomsen J.T.W. (2013). Environmental Burden of Disease Assessment: A Case Study in the United Arab Emirate	a national-scale Environmental Burden of Disease assessment framework for the UAE by estimating disease burden from multiple environmental risk factors such as air pollution, climate change, occupational exposure, water contamination, and food contamination.	Identify environmental risk factors → Identify climate/environment-sensitive diseases → Collect environmental exposure and health datasets → Calculate Relative Risk (RR) and Attributable Fraction (AF) → Estimate excess mortality and morbidity → Convert disease burden into DALYs/YLL/YLD → Perform uncertainty and sensitivity analysis → Develop integrated national EBoD estimates.	- PM2.5 and ozone data - Climate and temperature datasets - Mortality and hospital admission records - Occupational exposure data - Water and food contamination data - Greenhouse gas emission estimates - WHO and epidemiological RR coefficients	- Comparative Risk Assessment (CRA) - Relative Risk-based burden modeling - Attributable Fraction equations - Atmospheric modeling (CMAQ) - Scenario modeling - Sensitivity and uncertainty analysis	- Estimated mortality and healthcare burden attributable to multiple environmental exposures - Developed a modular EBoD framework usable for national policy planning - Identified air pollution and climate-related exposures as major contributors - Highlighted importance of uncertainty analysis in environmental-health studies	- closest frameworks to our EBoD architecture - it integrates multiple environmental risks into one burden framework. - We can reuse: • Relative Risk integration workflow • Attributable Fraction calculations • DALY/YLL/YLD conversion pipeline • uncertainty analysis methods • multi-exposure framework structure - They combine environmental monitoring with epidemiological evidence - Limitation: UAE environmental and demographic conditions are different from India, so RR thresholds and exposure assumptions may need to be adjusted for Indian populations.
Environment & Health (ACS Publications) Kang N., Li P., Xue T., Zhu T. (2024). *Development of a Method to Determine the Environmental Burden of Diseases and an Application to Identify Factors Driving Changes in the Number of PM2.5-Related Deaths*	To develop a comprehensive Environmental Burden of Disease (EBoD) framework and identify the factors driving PM2.5-related mortality changes in China between 2000–2010.	Identify PM2.5-related health outcomes → Collect exposure and demographic data → Apply exposure-response/Relative Risk (RR) functions using GEMM → Estimate attributable mortality burden → Decompose burden	- PM2.5 concentration data - Mortality statistics - Population and aging data - GDP/economic indicators - Urbanization statistics	- GEMM (Global Exposure Mortality Model) - Comparative Risk Assessment (CRA) - Mathematical decomposition equations - Nonlinear regression	- PM2.5-related deaths increased by ~620,000 between 2000–2010 - Population aging and increased exposure were major contributors - Economic growth and healthcare improvements reduced some burden - Showed environmental burden is influenced by both pollution and demographic transition	- explains *why* disease burden changes - The decomposition framework can help us separate the contribution of climate exposure, demographic change, and healthcare improvements in India. - We can reuse: • decomposition equations • RR integration methods • attributable burden calculations • demographic adjustment framework

in China between 2000 and 2010		into drivers such as aging, population growth, pollution increase, and economic development → Compare contribution of each factor to total mortality change.	- Healthcare infrastructure indicators - Census and county-level datasets - RR values from epidemiological studies	- Exposure-response epidemiological modeling		- Useful for building district/state-level burden estimation in India. - environmental burden studies become more policy-relevant when they identify the drivers behind increasing burden(they identified in this paper) - Limitation: mainly focused on PM2.5 mortality, while our work may include broader climate-sensitive diseases such as heat stress, respiratory illness, and vector-borne diseases.
World Health Organization (WHO), Environmental Burden of Disease Series Prüss-Üstün A, Kay D, Fewtrell L, Bartram J. *Assessing the environmental burden of disease at national and local levels: introduction and methods*. WHO Environmental Burden of Disease Series No. 1.	a standardized framework to estimate disease burden attributable to environmental risk factors including **air pollution**, **water contamination**, **occupational exposure**, **food contamination**, and other environmental determinants using DALY-based assessment.	Identify environmental risk factor → Define exposure distribution → Establish exposure-response relationship → Calculate attributable fraction (AF) → Estimate attributable mortality/morbidity → Convert to YLL + YLD → Compute DALYs → Aggregate national/local burden estimates.	• Exposure prevalence data • Relative Risk (RR) estimates • Mortality records • Disease incidence/prevalence • Disability weights • Life expectancy tables • Census population • Environmental monitoring data	• Comparative Risk Assessment (CRA) • Attributable Fraction equations • DALY framework • Exposure-response modeling • Scenario-based burden estimation	Demonstrated large disease burden attributable to modifiable environmental exposures. • Standardized DALY-based environmental risk attribution framework.	• Core methodological backbone for our India climate-health EBoD work. • AF and DALY computation pipeline directly reusable. • Important for structuring exposure → outcome → burden workflow. • Helps justify disease prioritization framework. • Limitation: climate-specific pathways are less detailed. • Useful for aligning methodology with WHO-standard burden estimation practices
Environmental Health Perspectives Campbell-Lendrum D, Woodruff R. *Comparative Risk Assessment of the Burden of Disease from Climate Change*.	Quantify future disease burden attributable to **climate change** through pathways involving **temperature-related mortality**, malnutrition, diarrheal disease, floods, and vector-related outcomes.	Define climate-sensitive health outcomes → Use climate projections → Estimate exposure changes under climate scenarios → Apply exposure-response functions → Calculate attributable	• Climate projections • Temperature data • Disease baseline incidence • Relative Risk coefficients • Population projections • Mortality estimates	• Comparative Risk Assessment (CRA) • Scenario-based modeling • Exposure-response modeling • Attributable burden estimation • Uncertainty analysis	• Climate change projected to substantially increase burden from diarrhea, malnutrition, and heat-related mortality. • Vulnerability highest in low-income regions. • Children disproportionately affected.	• relevant template for climate-health EBoD in India. • Useful for linking climate exposure pathways with disease endpoints. • Scenario-based burden estimation framework can be adapted for Indian climate projections. • Demonstrates how future climate risk can be embedded into DALY estimation • Limitation: coarse global-level assumptions; limited consideration of local adaptation.

		mortality/morbidity → Convert into DALYs → Compare across baseline and future scenarios.	• WHO disease burden datasets			
Journal of Public Health (OUP), 2017 Prüss-Ustün A et al. *Diseases due to unhealthy environments: an updated estimate of the global burden of disease attributable to environmental determinants of health.*	Update global estimates of disease burden attributable to unhealthy environmental conditions including **air pollution**, **water contamination**, sanitation deficits, chemical exposures, and occupational risks.	Identify environmentally attributable diseases → Assign attributable fractions using expert review and epidemiological evidence → Estimate attributable deaths and DALYs → Stratify by region, age, and disease category → Aggregate global burden estimates.	• WHO mortality databases • DALY estimates • Exposure prevalence • PM2.5 concentration •Water/sanitation indicators • Occupational exposure data • RR coefficients	• Comparative Risk Assessment (CRA) • Population Attributable Fraction (PAF) analysis • Epidemiological evidence synthesis • Burden attribution modeling	• ~23% of global deaths linked to environmental determinants. • Major contributors included **air pollution**, unsafe water, and occupational risks. • Children under 5 disproportionately impacted.	• Valuable for identifying priority environmental risk categories applicable to India. • Provides practical disease-environment linkage inventory useful for EBoD scope definition. • Reusable approach for selecting attributable fractions where direct local data are unavailable. • Limitation: some AF estimates derived from expert judgment rather than evidence.
Tropical Medicine and Health, 2025 *Disease burden attributable to high temperature between 1990 and 2021 in South Asia and Southeast Asia, with projections to 2045.*	Assess historical and projected disease burden attributable to **high temperature and temperature-related mortality** in South and Southeast Asia under future warming scenarios.	Collect historical temperature and disease burden data → Estimate temperature-health exposure relationships → Calculate attributable mortality and DALYs → Analyze temporal trends → Project future burden under warming scenarios to 2045	• Temperature data • Climate projections • DALY estimates • Mortality records • Population projections • Heat exposure metrics • Regional disease burden datasets	• Exposure-response modeling • Time-series burden analysis • Projection modeling • Attributable burden estimation • Trend analysis	• Heat-attributable disease burden increased substantially from 1990–2021. • Future warming expected to further elevate mortality and DALYs. • South Asia identified as highly vulnerable hotspot	• Highly relevant regional evidence for India-focused climate-health EBoD. • Provides South Asian burden estimates • Methodology is also useful for integrating heat exposure into Indian DALY calculations .• Projection framework valuable for future climate-health scenario analysis. • Could guide inclusion of heat-related diseases in national burden models. • Need to assess whether India-specific subnational granularity is available

Environmental Health, 2009 Knol AB et al. *Dealing with uncertainties in environmental burden of disease assessments*.	Develop structured approaches for handling uncertainties in EBoD assessments involving **air pollution**, environmental exposures, and risk attribution modeling.	Identify uncertainty sources → Categorize uncertainties (data, model, scenario, parameter) → Quantify uncertainty where possible → Apply sensitivity/uncertainty analysis → Transparently communicate uncertainty ranges in burden estimates. .	• Exposure estimates • RR coefficients • Model assumptions • Monte Carlo inputs • Epidemiological uncertainty ranges • Scenario assumptions	• Uncertainty analysis • Sensitivity analysis • Monte Carlo simulation • Probabilistic modeling • Structured uncertainty framework	• Uncertainty substantially influences EBoD estimates. • Transparent uncertainty reporting improves policy credibility. • Identified key uncertainty domains in environmental health modeling. .	•for tightening up our India EBoD estimates. • guidance for handling uncertainty in climate-health projections and RR estimates. • Monte Carlo and sensitivity-analysis concepts can strengthen robustness section of our work. • Particularly useful where Indian exposure-response evidence is sparse. • Helps prevent overconfidence in burden estimates and improves scientific defensibility for publication and policy use. .
Journal of the Epidemiology Foundation of India (JEFI), 2026 Ahir P., Bhati D.S., Yadav S. (2026). Air Pollution and Chronic Respiratory Diseases (CRDs) in Rajasthan: A Systematic Review of the Literatur	To systematically review the burden of chronic respiratory diseases associated with **air pollution**, **occupational exposure**, biomass fuel exposure, and particulate pollution in Rajasthan, and identify major respiratory outcomes, vulnerable groups, and research gaps.	Search multi-database literature using PRISMA guidelines → Screen and remove duplicates → Apply PECO-based inclusion/exclusion criteria → Extract exposure and respiratory outcome data → Assess study quality using AXIS/Newcastle-Ottawa tools → Perform narrative synthesis of CRD burden and exposure-response findings.	• PM2.5 / PM10 exposure data • Occupational silica dust exposure • Biomass fuel exposure • Spirometry measurements (FEV1, FVC, PEFR) •Mortality/morbidity records • Respiratory symptom prevalence •Tuberculosis/silicosis prevalence •Smoking/substance-use data • Exposure duration • Radiological finding	• Systematic Literature Review (SLR) • PRISMA methodology • Narrative synthesis • Cross-sectional and cohort evidence synthesis • Logistic regression • ANOVA • Exposure-response assessment • Risk factor analysis	• Silicosis prevalence ranged from 8%–52%. • Silico-TB prevalence ranged from 7%–12%. • Mining workers showed severe restrictive lung impairment and fibrosis. • Biomass and traffic-related pollution associated with reduced lung function. • Malnutrition significantly increased silicosis/TB risk (HR 5.61).	• Very relevant for identifying climate/environment-sensitive respiratory outcomes in India, especially for Rajasthan-like high-exposure settings. • Useful repository of exposure pathways linking **occupational exposure**, **PM exposure**, biomass fuel use, and respiratory disease burden. • The PECO framework, exposure categorization, and structured evidence synthesis approach can help in defining disease-selection criteria for our India climate-health EBoD work. • Strong regional evidence supporting inclusion of heat/dust/air-pollution-sensitive respiratory diseases in burden estimation. • Important insight: many studies lacked longitudinal exposure monitoring and robust uncertainty handling highlights a methodological gap we can improve upon in our study. • Could help justify subnational/state-level EBoD analysis for India instead of only national aggregation.

International Journal of Public Health Hennig F. et al. (2023). *Environmental Burden of Disease due to Emissions of Hard Coal and Lignite-Fired Power Plants in Germany, 2015.*	Estimate national disease burden attributable to long-term exposure from coal and lignite-fired power plants, specifically focusing on **PM2.5**, NO_2, and associated cardiovascular and respiratory outcomes linked to **air pollution** exposure.	Identify populations exposed to coal/lignite emissions → Model PM2.5 and NO_2 dispersion from power plants → Link pollutant exposure with cardiovascular and respiratory mortality using RR functions → Calculate attributable fractions (AF/PAF) → Estimate Years of Life Lost (YLL) attributable to emissions → Compare burden across hard-coal and lignite sources.	• PM2.5 concentration maps • NO_2 concentration estimates • Coal/lignite plant emission inventories • Population distribution data • Cause-specific mortality data • RR coefficients from epidemiological literature • National environmental monitoring datasets	• Comparative Risk Assessment (CRA) • RR-based attributable fraction modeling • Exposure dispersion modeling • YLL estimation • Cause-specific burden analysis	• PM2.5 emissions from lignite plants produced higher YLL burden compared to hard-coal plants. • NO_2-related burden was also substantially higher near lignite facilities. • Cardiovascular diseases were the dominant contributor to attributable burden. • Demonstrated measurable health burden from a single industrial emission source category.	• Structurally very similar to the UAE framework: emission modeling → exposure estimation → RR integration → AF → YLL/DALY estimation. • Helps us understand how environmental burden can be disaggregated by emission source rather than only by pollutant type. • The modular structure can later help us create separate India burden blocks for heat, coal pollution, occupational dust exposure, etc . • Also important from a policy perspective because it directly links industrial sources with measurable health loss, making findings easier for regulators and ministries to act upon.
Annual Review of Public Health Mannucci P.M. et al. (2025). *Global Burden of Disease from Environmental Factors.*	Quantify the global disease burden attributable to **air pollution**, **occupational exposure**, unsafe water, household pollution, and other environmental determinants using GBD 2021 estimates, with emphasis on climate-sensitive environmental risks.	Use GBD environmental risk-factor framework → Extract exposure-specific attributable deaths and DALYs from GBD 2021 → Stratify burden by environmental risk category → Compare contribution of each exposure to global mortality and DALYs → Discuss climate change amplification pathways affecting environmental burden.	• GBD 2021 DALY estimates • Exposure prevalence datasets • RR coefficients • WHO mortality datasets • Occupational exposure data • Household air pollution estimates • PM2.5 exposure estimates	• Comparative Risk Assessment (CRA) • Secondary GBD analysis • Population attributable burden analysis • Descriptive environmental risk-factor modeling	• Environmental and occupational risk factors contributed significantly to global deaths and DALYs in 2021. • **PM2.5** and household air pollution were among the largest contributors globally. • Climate change was identified as an amplifier of heat exposure, vector-borne disease transmission, flooding, and pollution-related burden.	• Important benchmark/reference paper for contextualizing India estimates against global environmental burden trends. • Strongly supports using GBD-style RR functions, TMREL assumptions, and CRA methodology in our India climate-health EBoD framework. • Useful for identifying which environmental exposures dominate globally and may need prioritization in India. • Also useful for discussion sections because it links environmental burden directly with climate change amplification pathways. • Helps provide international comparability and strengthens policy framing for India-focused climate-health burden estimation.

European Journal of Public Health (Supplement) Koopmans J.R. et al. (2019). *Introduction to the Environmental Burden of Disease concept.*	Explain the Environmental Burden of Disease (EBoD) concept and provide a practical framework for estimating disease burden attributable to environmental exposures such as **air pollution**, noise, and other environmental determinants.	Define EBoD concept → Estimate exposure prevalence in the population → Combine exposure with exposure-response/RR functions → Calculate attributable fraction (PAF) → Multiply attributable fraction with disease burden estimates → Estimate attributable deaths and DALYs for environmental exposures.	• Exposure prevalence datasets • RR coefficients • Disease incidence/prevalence data • Mortality records • DALY estimates • Environmental monitoring datasets	• Comparative Risk Assessment (CRA) • Population Attributable Fraction (PAF) framework • Conceptual methodological synthesis • Exposure-response burden estimation	• Clearly outlined the standardized EBoD workflow and key methodological requirements for environmental burden estimation. • Demonstrated how environmental and health datasets can be integrated through RR/PAF logic.	• Very useful conceptual/methodological reference for explaining our framework in a clean and understandable way to reviewers, professors, and policymakers. • Closely matches the exposure → RR → attributable burden → DALY flow we are trying to build. • Good paper for supporting methodology diagrams and explaining why RR/PAF approaches are central to EBoD estimation. • Especially useful because it simplifies the EBoD process into reproducible blocks rather than focusing only on results.
German Federal Environment Agency (UBA) German Federal Environment Agency (UBA). *Environmental burden of disease – national assessment concept and examples*	Demonstrate how a national environment agency operationalizes Environmental Burden of Disease estimation for **air pollution**, **heat exposure**, noise, and environmental health risks to support public-health policy and decision-making.	Link environmental exposure datasets with health datasets → Estimate population-level exposure distribution → Apply RR/exposure-response functions → Calculate attributable disease burden using PAF methods → Generate national EBoD estimates for environmental exposures → Translate results into policy-relevant indicators.	• National environmental monitoring datasets • PM2.5 exposure maps • Noise exposure maps • Mortality and disease incidence data • RR coefficients • National health datasets	• National-scale Comparative Risk Assessment (CRA) • RR-based attributable burden estimation • Environmental-health integration modeling • Policy-oriented EBoD framework	• Produced national-level burden estimates attributable to multiple environmental exposures in Germany. • Demonstrated direct application of EBoD outputs for environmental-health policymaking.	• Very valuable as a “mature EBoD system” example showing how environmental burden frameworks are institutionalized at national level. • Similar multi-exposure integration philosophy as the UAE paper . • Useful inspiration for how India could eventually integrate environmental-health surveillance with policymaking and ministry-level reporting. • Also demonstrates how EBoD outputs can be translated into actionab

Environmental Health Perspectives Patz J.A. et al. (2007). *Climate change and disability-adjusted life years*	Estimate DALY burden attributable to **climate change**, including impacts from **heat exposure**, malnutrition, diarrheal disease, flooding, and **vector-borne diseases** using Environmental Burden of Disease methodology.	Identify climate-sensitive diseases → Use climate-change scenario projections → Estimate changes in exposure patterns under future climate conditions → Apply RR/exposure-response relationships → Calculate attributable fractions → Multiply attributable fractions with baseline DALYs → Estimate climate-attributable disease burden under future scenarios.	• Climate scenario projections • Baseline disease incidence • DALY estimates • RR coefficients • WHO/GBD datasets •Exposure-response functions for climate-sensitive diseases	• Comparative Risk Assessment (CRA) • Climate-scenario modeling • RR-based PAF calculations • DALY estimation • Future burden projection analysis	• Demonstrated that climate-attributable DALYs can be estimated using standard EBoD methodology. • Highlighted substantial burden contributions from malnutrition, diarrheal diseases, flooding, and climate-sensitive infectious diseases. • Established one of the earlier DALY-focused climate-health burden estimation frameworks.	• Extremely relevant to our final India climate-health EBoD objective because this paper directly connects climate change exposure with DALY estimation. • Works very well alongside the Campbell-Lendrum climate burden paper as a foundational climate-EBoD reference. • Useful for understanding how future climate projections can be translated into attributable health burden estimates . • The scenario-based RR → AF → DALY framework is highly reusable for future India projections involving heat mortality, vector-borne diseases, and water-related outcomes. • Also useful for defending why climate-health impacts should be expressed in DALYs rather than only mortality counts.